High Validity, Low Transfer: Anchored Evaluation of Diffusion Counterfactual Explanations for Chest Radiographs

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Abstract

Visual counterfactual explanations are usually judged by Counterfactual Validity (CV): how often an edit flips the decision of the classifier that guided it. Because one model both steers and scores the edit, CV cannot separate a class-relevant change from an adversarial perturbation. We study this on chest radiographs with PPO-CF, a latent diffusion generator in which a PPO agent sets the sampling schedule around a frozen, LoRA-adapted backbone, and a soft background lock largely restricts edits to a dilated lung region. We pair CV with two anchored measurements: a Lesion Hit Rate against radiologist-drawn opacity boxes, and a held-out transfer ratio that re-scores counterfactuals with three classifiers never used for guidance. On RSNA Pneumonia, normal-to-pneumonia counterfactuals reach 94.5% CV, yet the held-out classifiers reproduce on average only 28% of the guidance classifier's score shift, and the one trained on NIH labels only 4%. Pneumonia-to-normal edits fall on annotated opacities 2.60× as often as their area predicts, against 2.06× with no learned policy, but a fixed schedule set to the policy's mean action recovers most of that gain, and edits stay diffuse within the lung in every configuration we train (21–31% of the change in the top 5% of pixels). Two mechanisms designed to localize edits, an annealed sparsity reward and a Grad-CAM constraint, both reduce lesion targeting; the latter collapses validity to under 2%. Judged by CV alone, our system would look far more successful than the anchored measurements indicate; anchored companion metrics should accompany CV whenever the guidance classifier computes it.

1. Introduction

Classifiers deployed on chest radiographs are typically audited with attribution maps, which indicate where evidence is concentrated but not what change would alter the prediction. Visual counterfactual explanations close this gap: given a query image x with prediction $f(x)$ and a target class c, a generator produces x_c that (i) is scored at c by the classifier, (ii) remains a realistic radiograph, and (iii) differs from x only in a minimal, class-relevant region, preserving anatomy, foreign objects and patient identity.

The standard success criterion, Counterfactual Validity (CV), is almost always computed with the same classifier whose gradients guided generation. It therefore cannot distinguish a class-relevant edit from an adversarial perturbation tuned to one model. This paper measures how much that matters for one system. We pair CV with two companion measurements, one anchored to human annotation (a Lesion Hit Rate against radiologist-drawn opacity boxes) and one to model independence (a held-out transfer ratio computed with classifiers that never guided generation), and apply them to a diffusion counterfactual generator we built. The concern is not hypothetical there: our system reaches 94.5% CV in the normal-to-pneumonia direction while the held-out classifiers reproduce on average 28% of the guidance shift. We have not re-evaluated other published generators, so we make no claim about how common this is; we claim only that nothing in the conventional metric suite would have surfaced it for ours.

The generator, PPO-CF, is the test case. It frames per-step guidance control as a sequential decision problem: a PPO agent sets the sampling schedule (strength, guidance scales, pass count, stopping) of a frozen, LoRA-adapted latent diffusion model, on the hypothesis that one hand-tuned schedule fits a heterogeneous test population poorly; FiRe [32] instead hand-designs adaptive guidance and early stopping (§2). The agent never touches the diffusion weights. The evaluation also tests that hypothesis, and largely refutes it. A fixed schedule set to the policy's mean action recovers most of the policy's gain over the generator's default schedule (§5.2).

Contributions.

1. An evaluation protocol that pairs CV with a lesion-anchored hit rate and a direction-specific held-out transfer ratio with a threshold analysis, together with the split-, seeding- and routing-level controls an arm comparison needs in order to mean anything (§4).
2. PPO-CF, a PPO-scheduled sampling controller around a frozen, LoRA-adapted Stable Diffusion generator with a soft, segmentation-gated background lock (§3), used as the case study.

3. Findings on that system (§5): high CV with low held-out transfer; lesion targeting above the default schedule but largely matched by a fixed schedule; two negative results for mechanisms designed to localize edits, one of them the largest effect we measure; and single-run arm comparisons in which seed-to-seed variation is of the same order as the smaller gaps between arms.

2. Related Work

Counterfactuals for chest radiographs. Singla *et al*. [26] introduced a GAN-based counterfactual generator for chest radiographs with the evaluation protocol (CV, FID and Foreign-Object Preservation, FOP) that became the de facto benchmark, which we adopt for comparability; FOP needs a detector trained on private annotations, so §4.1 uses a model-free substitute. Latent Shift [5] moves an autoencoder latent along the classifier gradient, RadEdit [17] edits radiographs with a diffusion model under multiple masks to stress-test classifiers, and FastDiME [31] accelerates diffusion counterfactuals with self-optimized masks, including on chest X-ray shortcuts. Diffusion counterfactuals for natural images [1, 11, 12] and in latent space (LDCE [8]) likewise score validity with the target classifier.

Evaluating beyond the guidance classifier. Re-scoring with a model not used for generation has precedents: DiME and ACE [11, 12] count non-target attribute changes with an independently trained oracle, and DVCE [1] uses adversarially robust classifiers so that gradient-guided edits become semantic. Our transfer ratio re-scores the *target* shift itself, as a per-direction ratio to the guidance shift on validators trained on other corpora, with a threshold analysis rather than one cut-off; the novelty is the pairing with CV and with a lesion anchor, not re-scoring as such. LHR adapts the energy-based pointing game of Score-CAM [28] (the share of a map's energy inside a box) from saliency maps to the counterfactual difference map, with an area null and a calibration (§4.2).

Diffusion editing and spatial locks. Classifier-guided diffusion [7, 16, 18] has displaced GANs for editable generation. We build on a LoRA-adapted [9] Stable Diffusion v1.5 [19] backbone with DDIM sampling [27]. RePaint [14] and Blended Latent Diffusion [2] re-composite the unedited region at every step; we adapt this as a soft lock through a dilated, feathered lung mask.

Localized and adaptive counterfactuals. ECED [15] edits a Score-CAM foreground and DocVCE [22] filters adversarial gradients and restores unneeded patches. FiRe [32] is closest in mechanism: it refines at a fixed noise level under saliency dual masks and *hand-designs* the controls PPO-CF learns, guidance that strengthens when more editing is needed and weakens near the target decision, and early stopping once the decision flips. PPO-CF instead learns per-step strength, guidance scales, pass count and stopping from a reward, around a lung rather than a saliency mask (a Grad-CAM mask collapsed our task, §5.5). Our finding that a fixed schedule at the policy's mean action recovers most of its gain (§5.2) suggests a hand-designed controller of FiRe's kind may suffice. None of these three methods reports a lesion-anchored metric or held-out target transfer, and none was run on our split and seeding, so we do not compare head-to-head (Supp. E3).

RL and schedules for diffusion. Most RL work on diffusion fine-tunes the weights against a reward (*e.g.* DDPO [3]). Sampling schedules have been optimized without RL, for noise levels (Align Your Steps [21]) and for guidance weights over time [30], where simple monotone schedules are hard to beat. Learning per-step sampling and guidance controls with RL around a frozen generator, for counterfactual editing, is less explored; in our setting the learned controls turn out close to a fixed schedule.

3. Method

3.1. Problem formulation

Given a frozen binary classifier $f : \mathcal{X} \to [0, 1]$ and a target class $c \in \{0, 1\}$, we seek x_c that trades off two objectives,

$$\min_{x_c} \Big(\underbrace{\mathrm{BCE}(f(x_c), c)}_{\text{classification}}, \underbrace{d(x, x_c)}_{\text{preservation}} \Big), \tag{1}$$

which we do not combine with a single fixed weight. The trade-off is set during sampling by the ratio of the classifier- to preservation-guidance scales, which the policy chooses per pass (§3.2, §3.3), and during policy training by the reward weights (Supp. Table R1). Here d is a masked distance penalizing change inside the segmented lung field and change to a frozen object-detector embedding (a foreign-object proxy), and leaving change outside the lung field free, subject to the soft background lock below, which largely removes that freedom away from the lung field.

3.2. Latent generator and background lock

Images are encoded to a $4 \times 32 \times 32$ latent by a frozen SD1.5 VAE. A lightweight MLP maps $(c, \text{pooled latent})$ to a single cross-attention conditioning token, replacing SD1.5's text conditioning with a scalar class signal. Rank-8 LoRA adapters are inserted into the UNet's attention projections and trained on the standard ϵ-prediction objective on the RSNA training fold (§4.4), adapting the natural-image prior toward radiography without touching the VAE or the bulk of the UNet.

Guided DDIM sampling. Generation runs a partial, SDEdit-style forward-then-reverse DDIM trajectory whose length is a strength fraction of $T_{\text{diff}} = 50$ steps. At each step the UNet's noise prediction gives both the predicted clean latent $\hat{z}_0$ and the DDIM update z_{t-1}. Two guidance gradients are taken with respect to $\hat{z}_0$, through the VAE decoder D, and added to z_{t-1}:

$$z_{t-1} \leftarrow z_{t-1} - w(i)\, m \odot \left(\alpha_{\text{cls}} \nabla_{\hat{z}_0} \mathcal{L}_{\text{cls}} + \alpha_{\text{pres}} \nabla_{\hat{z}_0} \mathcal{L}_{\text{pres}}\right), \tag{2}$$

where $\mathcal{L}_{\text{cls}} = \text{BCE}(f(D(\hat{z}_0)), c)$, $\mathcal{L}_{\text{pres}} = \mathcal{L}_1^{\text{seg}} + \eta\, \text{MSE}_{\text{obj}}$, $\mathcal{L}_1^{\text{seg}}$ is the lung-mask-weighted L_1 distance from the original image, MSE_{obj} compares object-detector features, and $\alpha_{\text{cls}}, \alpha_{\text{pres}}$ include the per-pass scales set by the policy (values in Supp. Table R4). The update is gated by the soft edit mask m defined below and by a linear ramp $w(i)$ rising from 0 at a learned fraction of the trajectory, which avoids the discontinuity of a hard guidance switch.

Background lock. After each guided step the latent is blended toward an anchor: the original image's latent forward-noised to the next timestep, and used clean at the final step,

$$z_{t-1} \leftarrow m \odot z_{t-1} + (1 - m) \odot \tilde{z}_{t-1}^{\text{orig}}. \tag{3}$$

Here $m \in [0, 1]$ is the soft edit mask: the original image's lung-probability map, resized to the 32×32 latent grid, dilated by two latent cells (≈ 16 px) and Gaussian-feathered (σ of one cell). It is not thresholded. The blend does not depend on how the guidance gradient behaves: where $m \approx 0$, far from the lungs, the final latent is essentially the original image's, rather than being repainted by the free SDEdit process, which is what an unconstrained high-strength schedule does to the entire thorax. The constraint is soft rather than exact. The dilated, feathered band around the lungs remains editable, and because the VAE decoder mixes neighbouring latent cells, preservation in pixel space outside the mask is approximate.

3.3. PPO-scheduled sampling

Rather than fixing strength, guidance scales and pass count, a PPO agent [23] emits a 9-dimensional continuous action at each of up to $T = 25$ policy steps, mapped through sigmoid transforms to the controls in Table 1. Each policy step runs two to four chained generator passes (two, plus one for each of a_4, a_5 thresholded at $\sigma(a) > 0.5$, identically in training and evaluation), with the exact mapping and per-pass schedule modifiers given in Supp. Sec. R2.

Observation and architecture. The observation is a flat vector of $8{,}192 + 65{,}536 + 3 = 73{,}731$ entries: the VAE latents of the original and current images (two $4 \times 32 \times 32$ latents, 8,192 entries; drawn from the VAE posterior rather than taken at its mean, with the current-image latent redrawn at every step, so the observation is stochastic even for a fixed image), the soft lung-segmentation mask of the current image at the 256×256 image resolution (65,536 entries, i.e. 88.9% of the vector), and three scalars: the step fraction t/T, the current distance $d_t = |f(x_t) - c|$ and the target c. The agent is a policy network with a shared trunk and policy, value and three auxiliary heads (feasibility, progress, stop); the trunk consumes the full flattened observation, and the policy head outputs nine Gaussian means with a state-independent, per-dimension learnable log-std. It is trained with the clipped PPO objective and GAE (hyperparameters in Supp. Table R3); episodes end when $d_t < \epsilon_{\text{stop}} = 0.08$ or after $T = 25$ steps.

Table 1. Action space. All nine dimensions are emitted at every policy step by a policy conditioned on the image state and step index, so it can emit image- and step-dependent controls. In an evaluation replay the n→p policy holds a_0, a_4, a_5, a_7 at or near their bounds and a_6 near-constant, varying a_1–a_3, a_8; the p→n policy varies all nine (§6). Because a_7 sits within 1% of its upper bound on 97.5% of n→p steps, that bound may be binding; we did not train with a wider range. The a_0 range is the base strength s; pass k of a step applies $\min(0.9,\ s\beta(1+0.15k))$, with β from a_6, so applied strengths span $(0.17, 0.9]$. †Narrowed from $[0.5, 3.0]$ in an earlier code version (Supp. Table R5).

	Controls	Range
a_0	DDIM strength	$[0.20, 0.70]$
a_1	classifier-guidance scale	$[0.3, 1.2]^{\dagger}$
a_2	preservation-guidance scale	$[0.3, 1.5]$
a_3	stop signal τ_t	$[0, 1]$
a_4, a_5	extra pass, thresholded at $\sigma(a) > 0.5$	$\{0, 1\}$
a_6	per-pass strength boost	$[0.85, 1.15]$
a_7	guidance bias (cls vs. pres)	$[0.70, 1.30]$
a_8	guidance ramp-up start fraction	$[0.1, 0.5]$

Reward. The per-step reward is a weighted sum of: progress $d_t - d_{t+1}$; two distance penalties, $-d_{t+1}$ and $-d_{t+1}^2$; an L_1 fidelity penalty against the original image; a lung-mask-weighted L_1 preservation penalty plus a KL term between the foreign-object detector's softmaxed features on the original and current images; a constant per-step cost; a validity bonus when $d_{t+1} < \epsilon_{\text{stop}}$ (which also ends the episode); and a stop bonus when, in addition, $\tau_t > 0.5$. The sparsity arm adds an annealed L_1 term. The whole n→p reward is multiplied by 2.25. Weights and the implemented form of each term are listed in Supp. Table R1; Supp. Sec. R collects every setting needed to reproduce the method.

3.4. Directional policies

The environment is built around a fixed target, so one policy is trained for n→p ($c = 1$) and another for p→n ($c = 0$). Each direction fine-tunes its own LoRA adapter from the same data, seed and recipe, and the resulting weights differ (relative difference 0.385 in the default arm, 33 to 259 times that of the other arms; Supp. R4). The default and Grad-CAM arms are evaluated with each direction's own adapter. Through a loading default, the single-classifier, 3-model and sparsity arms used the n→p adapter for both directions, so where they are compared with the default we use the default evaluated under that same protocol (Table 4); its p→n LHR lift is then 0.077 higher (95% CI $[-0.001, +0.159]$), not separable from the $+0.016$ ($[-0.045, +0.098]$) between two reruns of one protocol on different GPU types. Separate policies also suit an asymmetry we observe throughout: synthesizing a plausible, spatially coherent opacity pattern in a clean lung field is a harder and more under-constrained generative problem than removing one, which is closer to an inpainting operation the pretrained prior already performs well.

4. Evaluation Protocol

4.1. Conventional metrics

Following [26], the evaluation set is split by classifier confidence into $X_n = \{x : f(x) < 0.2\}$ and $X_p = \{x : f(x) > 0.8\}$, up to $n_{\text{eval}} = 200$ each, and each directional policy generates from the corresponding group with its deterministic (mean) action. Full definitions of every metric below are given in Supp. Sec. R6.

Counterfactual Validity. With $s = 2c - 1$ ($+1$ for n→p, -1 for p→n),

$$\text{CV} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\, s\, \Delta_i \geq \tau \,\wedge\, B_i \,], \tag{4}$$

where $\Delta_i = f(x_c^i) - f(x^i)$, B_i requires $s\,(f(x_c^i) - 0.5) > 0$ and $s\,(f(x^i) - 0.5) \leq 0$, f is the guidance classifier (the ensemble mean in ensemble arms) and $\tau = 0.8$. On X_n and X_p the crossing condition B_i is redundant: $f(x) < 0.2$ and $s\,\Delta_i \geq 0.8$ already give $f(x_c) \geq 0.8$ for n→p, and symmetrically for p→n. The tightening over the original, direction-agnostic definition therefore comes from the sign requirement alone: an image moving the wrong way by a large margin is no longer credited.

FID. Between each direction's counterfactuals and the real images of its target group (X_p for n→p, X_n for p→n; selected by f, not by label), on 1024-d features of a TorchXRayVision DenseNet-121 [6] with `rsna` weights rather than Inception-v3. These weights also guide generation in every trained arm (the 3-model ensemble is `rsna`, `chex`, `mimic_nb`), so FID is not independent of guidance. With $n \leq 200$ samples in a 1024-d feature space FID is high-variance; we report its bootstrap interval and KID in Supp. E6 and do not rank arms by it.

Identity Preservation (IP). As a dependency-free substitute for FOP we report MS-SSIM and HF-corr $= (1 + r)/2$, where r is the Pearson correlation between the high-pass-filtered source and counterfactual (0.5 means uncorrelated), a proxy for whether thin structures such as tubes, leads and pacemakers survive editing.

Edit concentration. The share of $\sum |x_c - x|$ held by the top 5% of pixels, averaged over pairs; spatially uniform change scores exactly 0.05. We also report the fraction of pixels with $|x_c - x| > 0.01$; part of it is VAE reconstruction error, separated out in Supp. E6. There is no calibrated threshold; as an illustrative heuristic only, a compact lesion-shaped edit would place roughly half or more of its mass in the top 5%.

4.2. Ground-truth-anchored Lesion Hit Rate

CV, FID and IP share a blind spot: none of them can tell whether the edit landed on the pathology. RSNA ships 9,555 radiologist-drawn opacity boxes [25], which we retain. LHR adapts the energy-based pointing game [28]: it measures the fraction of the difference map $|x_c - x|$ falling inside the annotation, per image, against the box's own area fraction as the null, and reports the mean per-image lift (hit rate over null). What is new relative to the pointing game is the input (an edit rather than a saliency map), the area null, and the calibration below. LHR is defined only for p→n, since n→p sources are classifier-negative and carry no annotation. Details that the definition leaves open are listed in Supp. Sec. R6.

We calibrate the metric before use on synthetic edits with known ground truth: a uniform wash scores lift 1.00 ($p = 0.59$, chance, as it should), an on-lesion edit scores 12.4 ($p = 3 \times 10^{-49}$), an off-lesion edit scores 0.

4.3. Held-out transfer

Because CV is computed with the guidance classifier, it cannot separate a class-relevant edit from an adversarial perturbation. We therefore re-score every counterfactual with three TorchXRayVision DenseNet-121 classifiers that never participated in guidance and report, per validator v and direction, the ratio of the validator's mean signed score shift to the guidance model's own shift over the same N images,

$$\rho_v = \frac{\sum_i s\,\big(f_v(x_c^i) - f_v(x^i)\big)}{\sum_i s\,\big(f(x_c^i) - f(x^i)\big)}, \tag{5}$$

a ratio of means rather than a mean of per-image ratios, undefined when the guidance shift is zero. 95% intervals are percentile bootstraps over the N images (10,000 resamples, numerator and denominator resampled jointly). The validators differ in how independent they are of RSNA. `pc` (PadChest [4]) and `mimic_ch` (MIMIC-CXR [13]) share no images with it. `nih` was trained on NIH ChestX-ray14 [29], of which RSNA's images are a subset [25], so it has probably seen our test fold's source images, though with NLP-mined labels rather than RSNA's radiologist labels, and its Pneumonia label is rare (≈1% prevalence). We therefore lead with the three-validator mean and report `nih` separately as the cell with the lowest transfer, not as the most independent one. Of the guidance models, `rsna` was trained on RSNA data and `chex` on CheXpert [10]. The denominator must be the shift of the classifier(s) that actually guided generation, so it is read from the configuration stamp each evaluation shard carries, which records the guidance classifiers for the default and later arms (the single-classifier arm's shards predate this key and are treated as single-classifier guidance). A ratio below 0.15 we treat as the adversarial regime; this is a heuristic operating point, not a calibrated boundary (§5.3). A ratio of 0.5, meaning validators reproduce half the guidance shift, is used only as an illustration and not as a threshold.

4.4. Controls the comparison requires

A control-arm comparison needs three preconditions that we had assumed and that did not hold (full account in Supp. Sec. E1): (i) one deterministic, persisted, patient-level split shared by LoRA fine-tuning, the PPO pool and evaluation, replacing a shared dataset object under which evaluation images had been seen during fine-tuning; (ii) seeded training, since two identical unseeded submissions produced two different policies; and (iii) control arms exercised end to end, which surfaced a routing bug and a shared output directory that would each have produced a well-formed but wrong table. Every reported run uses the 80/20 patient-level split with seed 0, LoRA seed 0, PPO seeds 1 (n→p) and 2 (p→n), and evaluation base seed 0. Generator fine-tuning, PPO rollouts and the validity probe use only the 80% training fold; every evaluated image is in the 20% test fold. Every default-arm number in the paper comes from one evaluation in which each direction uses its own adapter, on the same 400 test images and per-image seeds as the no-policy, fixed-schedule and second-seed runs of §5.2 and §5.4. The single-classifier arm shares split, seeds, reward weights, action ranges and anchoring with the default arm and differs only in the guidance flag, and through it in the classifier-selected groups (78 of 200 p→n sources shared) and in hardware.

5. Experiments

5.1. Setup

We use the RSNA Pneumonia Detection Challenge dataset [20] (DICOM, min–max normalized, resized to 256×256, binary label collapsed from bounding-box annotations). In the default configuration the guidance classifier is the mean-probability ensemble of two TorchXRayVision DenseNet-121 models (`densenet121-res224-rsna`, `densenet121-res224-chex`); the single-classifier arm uses `rsna` alone; segmentation uses TorchXRayVision PSPNet for lung and heart channels; a YOLOv8n feature extractor supplies the foreign-object embedding. Each policy is trained for 800 episodes per direction; for the 2-model ensemble run the two directions took 8h40m (n→p, on an A100) and 11h00m (p→n, on an H100), both including LoRA fine-tuning (5.7 h and 5.1 h).

5.2. Main results

Table 2 reports the default configuration (2-model guidance ensemble, no Grad-CAM mask, no sparsity reward). The policy concentrates edit mass on radiologist-annotated lesions far above chance (2.60×), and adds to that over the generator's default schedule without a policy, run on the same images with the same guidance, groups, adapters and seeds: lift 2.064 (paired gap $+0.532$, 95% CI $[+0.423, +0.647]$, $d = 0.764$, better on 128/148 images) and p→n CV 0.225 (90 images gain validity, 1 loses it; McNemar $p = 7.4 \times 10^{-26}$). In n→p it adds nothing measurable (CV 0.945 vs. 0.940, 4 vs. 3 discordant). And yet the edits are diffuse within the lung. About 82% of the difference mass falls inside the editable region (49–59% of the image), but only a quarter to a third lies in the top 5% of pixels, against 0.05 for uniform change. The 65% of pixels changed is inflated by the VAE round trip, which alone changes about half of them; measured against $D(E(x))$ it is 41–44% (Supp. E6). The no-policy schedule scores lower FID and KID than the policy in both directions (FID bootstrap intervals do not overlap), but FID cannot rank arms: its extractor also guides generation. Clinical proxies support no clinical claim (Supp. E2).

A fixed schedule matches the policy. To ask whether the gain comes from per-image adaptation, we hold every control at the policy's mean action over its evaluation steps, with the same step budget and validity-based stopping and no training. On the same images this fixed schedule recovers 87% (95% CI $[71, 102]$%) of the policy's LHR gain over the no-policy schedule and 98% ($[86, 111]$%) of its p→n CV gain, and it is not distinguishable from the policy (LHR -0.071, $[-0.161, +0.019]$, $p = 0.12$; p→n CV 0.660 vs. 0.670, 14 vs. 16 discordant). In n→p it is valid on exactly the same 189 images. Its mean p→n transfer is lower

Table 2. Default configuration, $n = 200$ per group, from one evaluation in which each direction uses its own generator adapter (§4.4). LHR is defined only for p→n.

Metric		n→p	p→n
CV	fraction valid, τ=0.8	0.945	0.670
	mean	0.808	
LHR	lift vs. annotation-area null	—	2.595
	$t = 24.09$, $p = 6.6 \times 10^{-53}$, $n = 148$		
Transfer	mean over 3 validators	0.281	0.276
EditConc.	top-5% mass (uniform=0.05)	0.254	0.314
	fraction of pixels changed	0.647	0.634
FID	vs. real target class	22.38	4.76
IP	MS-SSIM	0.924	0.868
	HF-corr	0.948	0.913

Table 3. Held-out transfer ratio, single- vs. ensemble-guided trained policies, $n = 200$ per group. Values below 0.15 are in the adversarial regime. Ensemble row from the evaluation of Table 2; single-classifier row from a recomputation on the saved counterfactuals (Supp. R5).

	nih		pc		mimic_ch	
Policy	n→p	p→n	n→p	p→n	n→p	p→n
Single-classifier	0.081	0.100	0.414	0.224	0.197	0.112
2-model ensemble	0.044	0.165	0.465	0.306	0.333	0.357

(0.244 vs. 0.276). What the policy contributed here is an aggressive schedule, not per-image adaptation. A fixed schedule picked from a grid by the training reward, without RL (Supp. E5), does not reach the policy: its p→n CV is comparable (0.720 vs. 0.670, $p = 0.12$), but its LHR lift is lower (-0.187, $[-0.279, -0.095]$) and so is its transfer (0.211).

5.3. CV overstates what the edits achieve

Table 3 is consistent with the claim that CV overstates what the edits achieve, but is not decisive. Against the earlier single-classifier policy, the 2-model ensemble-guided policy has lower mean CV (0.808 vs. 0.940) but higher LHR lift (2.595 vs. 2.293) and higher 3-validator mean held-out transfer in both directions (Table 4; same orderings under matched adapters); the one per-validator exception, `nih` n→p, is not distinguishable from noise (below). The metric that fell is the one that shares a classifier with the generator. The two runs differ in guidance, in the classifier-selected groups and in hardware (§4.4), and with one training run per arm these differences are not separated from run-to-run variation (§5.4).

A single-run point estimate cannot resolve the apparent `nih` regression, so we bootstrap it (§4.3). For n→p, $[0.048, 0.114]$ (single) vs. $[0.005, 0.082]$ (ensemble): the intervals overlap substantially and the apparent worsening is not distinguishable from noise. Both sit entirely below

Table 4. Five trained-and-evaluated arms. Bold marks the default. Transfer is the mean over {nih, pc, mimic_ch}; *over {nih, pc} only (leak-free set), not comparable to the other rows. $n = 200$ per group, except the 3-model arm's p→n group ($n = 32$, 27 annotated). Paired LHR tests compare each arm with the default under the same adapter protocol: ‡against the default evaluated with the n→p adapter for both directions, as those arms were (LHR 2.673, transfer 0.282/0.292 under that protocol); Grad-CAM, whose two adapters are identical, against the per-direction evaluation.

Arm	LHR	Transfer n→p/p→n	LHR vs. default (paired)
Single-classifier	2.293	0.230/0.145	worse (d=0.482)‡
2-ens (default)	**2.595**	**0.281/0.276**	—
3-model ensemble	2.492	0.350/0.276*	unresolved (n=12)‡
Sparsity reward	2.398	0.311/0.212	worse (d=0.512)‡
Grad-CAM constraint	1.636	0.161/0.103	much worse (d=1.068)

0.15, so what is established is that `nih` n→p stays confidently low under either policy. For p→n, $[0.075, 0.126]$ vs. $[0.138, 0.193]$: the intervals do not overlap (resampling within a single training run). We report each policy's interval separately rather than a difference-of-ratios bootstrap, because the classifier that sorts images into X_n/X_p changed between runs and the two groups share only part of their members; a paired resample would synthesize a pairing that is not there.

The 0.15 cut-off is a conservative operating point, not a calibrated boundary, so we recount Table 3's adversarial cells at other thresholds (point estimates, strictly below the threshold). `nih` n→p is adversarial for both policies at every threshold from 0.10 to 0.25. Of the six cells, the single-classifier policy has 1/3/4/5 adversarial at 0.10/0.15/0.20/0.25 and the ensemble 1/1/2/2. Two cells sit on boundaries: the single-classifier `nih` p→n ratio is 0.100 and so is not strictly below 0.10, and the ensemble's `nih` p→n ratio of 0.165 crosses at 0.20, so reading it as non-adversarial holds only for thresholds up to 0.165.

This is the paper's central empirical claim. In n→p the default policy reaches a CV of 0.945, yet the three held-out validators reproduce on average 28% of the guidance shift and `nih` 4%; the single-classifier policy shows the same pattern (CV 0.980, mean transfer 0.230, `nih` 0.081). In that direction the policies largely produce perturbations that other models do not see, rather than explanations of the pathology, and nothing in the conventional metric suite surfaces that.

5.4. Five trained arms

Every arm in Table 4 is a policy trained for 800 episodes per direction and evaluated end to end, with one training run per arm. All five share the split and seeds of §4.4. LHR verdicts are paired tests on the images two arms share, never differences of group means; transfer comparisons use each

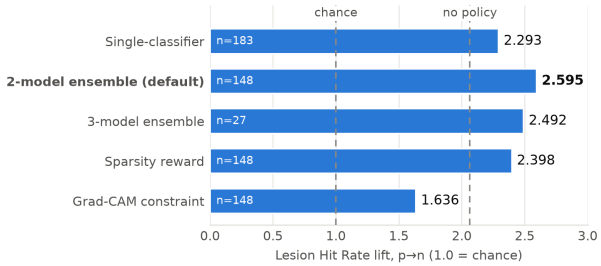


Figure 1. LHR lift (p→n) of the five trained arms (Table 4). Dashed lines mark chance (1.0) and the no-policy schedule run on the default's images, guidance, adapters and seeds (2.064).

arm's own bootstrap CI (§5.3).

Ensemble vs. single-classifier guidance, and seed variation. The two groups share 78 of 200 p→n sources, 76 of them annotated (two shared sources carry no box). On those 76 images the paired LHR test favours the ensemble-guided policy ($+0.285$, 95% CI $[+0.155, +0.418]$, $d = 0.482$ under the matched adapter protocol; $+0.189$, $d = 0.316$ against the per-direction evaluation). A second PPO seed of the default configuration, whose two directions share one adapter and were both resumed from checkpoints without saved random-number state, scores lower on the same 400 images: LHR lift 2.479 (paired -0.117, $[-0.227, -0.006]$) and p→n CV 0.495 vs. 0.670. Seed-to-seed variation is thus of the same order as the single-vs.-ensemble gap, which we read as unreplicated. Without a policy (groups selected by label), ensemble guidance raises paired LHR lift over `rsna`-only guidance by $+0.069$, $+0.069$ and $+0.058$ across three evaluation seeds (all 95% CIs excluding zero), a smaller effect that is not a replication of the trained comparison. Growing the ensemble to three members improves `pc` n→p transfer from 0.465 to 0.623 with non-overlapping CIs, does not resolve on LHR ($n = 12$), and lowers mean CV to 0.624.

The single-classifier policy beats a random one, but barely. On its 183 annotated images, the single-classifier policy beats the random-action arm by $+0.090$ (95% CI $[+0.021, +0.160]$, $p = 0.012$, $d = 0.189$, better on 105/183). The effect is real but small, and smaller than either negative result below.

5.5. Two negative results

An annealed sparsity reward fails at its own objective. Edit concentration stays in the 0.21–0.31 band in every arm we measure. The sparsity term was built to address this and had been disabled throughout, leaving its annealing ramp inert. We enabled it and retrained both directions. In n→p the validity safeguard (Supp. Sec. R1) froze $\lambda_{\text{sparsity}}$ at 0 before its ramp began, so only p→n was trained with the term; the n→p changes are run-to-run variation (CV 0.935 vs. 0.945). In p→n, concentration moved by less than 0.02. Meanwhile, under the matched adapter protocol, a paired test on the 148 shared annotated images gives a lift gap of $+0.275$ favouring the arm *without* it (95% CI $[+0.188, +0.363]$, $p = 4.7 \times 10^{-9}$, $d = 0.512$), and p→n transfer falls from 0.292 to 0.212 with non-overlapping CIs on two of three validators. Without a working spatial mask the term degrades to plain unweighted L_1, which discourages edit mass uniformly rather than off-target mass specifically. This is a plausible account of why it competes with precise editing while failing at its own goal.

A Grad-CAM spatial constraint collapses the base task. In a policy-free sweep, constraining edits to a Grad-CAM [24] region lowered LHR lift rather than raising it, so we diagnosed it directly. Scoring 263 member-image observations under the exact ensemble Grad-CAM path used in training contradicted the explanation on record: it is `chex`, not `rsna`, that degenerates (66/263, 25.1% vs. 1/263), and it does so at *low* predicted probability rather than near 0.5, consistent with output saturation zeroing the gradient. When `chex` degenerates the averaged mask falls back toward the other member alone, and CAM–lesion overlap falls with it (2.052, $n = 127$ on clean episodes vs. 1.347, $n = 23$ on partially degenerate ones). The remedy on record, a finer target layer, eliminates the degeneracy (0%/0.4%) but drops baseline CAM–lesion lift to 1.196, barely above chance.

A policy trained under that remedy moved in the same direction, by a larger margin: CV collapses from 0.808 to 0.0175 (7 of 400 counterfactuals valid), the paired LHR gap is $+0.960$ favouring the unconstrained arm (95% CI $[+0.815, +1.108]$, $p = 3.6 \times 10^{-26}$, $d = 1.068$, better on 138/148 images), and transfer falls on all three validators. This is the largest paired effect we measure, against a mechanism introduced specifically to help: constraining edits to a finer, less confident CAM region makes the base task substantially harder, and every dependent metric degrades with it.

5.6. Qualitative examples

In Fig. 2 the guidance score crosses the CV margin ($f(x) = 0.086 \rightarrow f(x_c) = 0.995$ for n→p; $0.848 \rightarrow 0.017$ for p→n). The p→n edit is more concentrated than the n→p one, but both remain diffuse relative to a lesion-shaped edit (§5.2).

6. Limitations

- **One dataset, one system.** Every result is on RSNA Pneumonia with one generator family. We did not re-

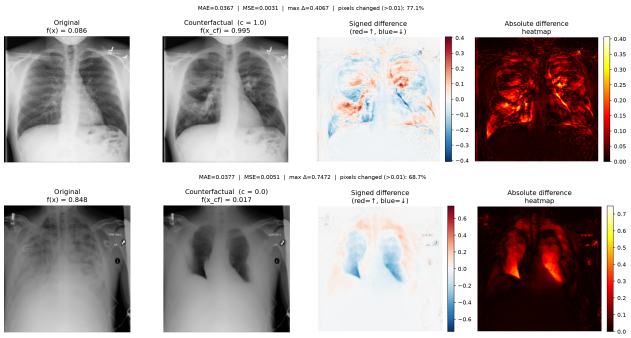


Figure 2. n→p (top) and p→n (bottom), default configuration, policy mean action. Original, counterfactual, signed and absolute difference. Change concentrates over the lung fields; anatomy and external hardware outside the lung mask remain visible, with smaller residual changes (including at the hardware) consistent with a soft rather than exact lock.

evaluate other counterfactual methods under our protocol (Supp. E3), so the gap between CV and the anchored metrics is established for our system only.

- **Edits remain diffuse.** No arm we trained moved edit concentration meaningfully; every one sits in the same 0.21–0.31 band. The lock keeps at least 93% of the change relative to the VAE round trip inside the editable lung region (Supp. E6), so the diffuseness is within the lung. Whether diffuseness is fixable within a gradient-guidance approach is open.
- **n→p transfer is low.** The three validators reproduce on average 28% of the guidance shift, and `nih` stays below every threshold from 0.10 to 0.25 under both policies; the 0.15 threshold itself is not formally calibrated (§5.3).
- **Classifier overlap with the evaluation data.** The `rsna` guidance model and the FID extractor were trained on RSNA data, and `nih` on NIH ChestX-ray14, which contains RSNA's images (§4.3); these models may have seen our test images, which can inflate CV and the `nih` ratio. `chex`, `pc` and `mimic_ch` have no RSNA exposure. Generator fine-tuning, PPO rollouts and the validity probe use only the 80% training fold.
- **One training run per arm.** A second PPO seed of the default configuration differs from the first by amounts of the same order as the single-vs.-ensemble gap (LHR lift -0.117 vs. $+0.189$; §5.4), and it is itself imperfect (shared adapter; both directions resumed from checkpoints without saved random-number state); no trained-arm gap has a proper run-to-run error bar. The single-classifier and ensemble runs also differ in classifier-selected groups (78 of 200 p→n sources shared) and hardware. Three arms were evaluated with the n→p adapter for both directions, which we match in their comparisons (§3.4); the Grad-CAM arm's n→p policy was trained for updates 1 to 49 under a different adapter from the one used at evaluation.
- **Adaptivity is limited** (action statistics from an evaluation replay with the policy's mean action, not read from the saved counterfactuals). In n→p, 199 of 200 episodes end after one step; strength sits at its lower bound and the pass count is three on every step, guidance bias is within 1% of its upper bound on 97.5% of steps, and strength boost is near-constant, while the classifier and preservation scales, stop signal and ramp start vary between images. In p→n the median episode is one step and the mean 5.46, 31 episodes reach the 25-step cap, and over the 71 multi-step episodes 67 to 78% of the action variance is between images. A fixed schedule at the policy's mean action matches the policy (§5.2), so per-image adaptation adds nothing we can detect, and a grid-tuned fixed schedule targets lesions less. The 37.9M-parameter policy was trained on at most 800×25 transitions per direction, far fewer given the short episodes.
- **Sample size and proxies.** The 3-model arm's p→n group is small ($n = 32$, 27 annotated; 12 shared for LHR). FID at $n \leq 200$ in a 1024-d space is high-variance (Supp. E6). FOP is not computable offline, so IP measures a related property, and SCP is degenerate in this configuration.
- **Not measured:** convergence curves (two logged points per direction); a LoRA-rank ablation; Score-CAM or consensus masks; a wider a_7 range; n→p lesion targeting (LHR needs annotated sources). The lock width was not tuned (exploratory check in Supp. E4).

7. Conclusion

We evaluated a PPO-scheduled, LoRA-adapted latent diffusion counterfactual generator for chest radiographs under a protocol that pairs classifier-scored validity with a lesion-annotation-anchored hit rate and a held-out transfer ratio. Under CV alone the system looks strong: 94.5% of normal-to-pneumonia edits flip the guidance classifier. The anchored metrics qualify that in three ways. Held-out classifiers reproduce on average 28% of the guidance shift in that direction. Pneumonia-to-normal edits do land on annotated opacities (2.60× the area null, against 2.06× without a policy), but a fixed schedule at the policy's mean action recovers most of that gain, so the learned schedule is close to a fixed, aggressive one. And edits stay diffuse in every configuration we trained; the two mechanisms designed to localize them both reduced lesion targeting, one by the largest effect in the study. Our evidence is one system on one dataset, with one training run per arm. Within it, CV computed with the guidance classifier would have overstated what the edits achieve, and an anchored companion metric (to human annotation, to model independence, or both) costs little and changes the conclusion.

LLM Usage Statement

We used an LLM-based coding and writing assistant (Claude, Anthropic) in two ways. (i) Code: it helped write and debug parts of the experiment and analysis code, including evaluation, bootstrap and paired-test scripts and job-submission scripts, and it helped trace reported values back to the logs and result files they came from. (ii) Writing: it helped edit and restructure the manuscript text, tables and captions. The authors designed the study and the method, chose the experiments, ran all experiments, and checked every number and claim in the paper against the saved outputs. No LLM was used to generate data, images or annotations, and no LLM is part of the method.

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